# CHARAN REDDY YAKKANTI

Charlotte, NC 28027 • (704) 652-8170 • charanreddy.business@gmail.com
linkedin.com/in/charan-reddy-yakkanti-966712321

**Curriculum Vitae**

## PROFILE

Undergraduate dual-degree candidate in Computer Science and Finance at the University of North Carolina at Charlotte. Current work is in multi-agent systems: the coordination layer that allows several autonomous processes to pursue a shared objective without duplicating effort or acting on stale state. Prior work spans autonomous vehicle control, combinatorial optimization and applied deep learning, alongside client-facing experience in technical recruitment.

## EDUCATION

### University of North Carolina at Charlotte

*Charlotte, North Carolina | Expected May 2028*

Bachelor of Science, Computer Science

Bachelor of Science, Finance

Cumulative GPA: 3.6 / 4.0

**Relevant coursework:** Data Structures and Algorithms; Corporate Finance; Professional Selling (personal selling process, consultative and social selling).

## PROFESSIONAL EXPERIENCE

### VibeGrow • Product Engineer

*September 2026 to Present | Remote*

VibeGrow is a pre-launch startup developing agent swarms intended to start and operate businesses autonomously, covering market research, product development, customer acquisition and daily operations.

- Build multi-agent coordination systems, with a focus on orchestration logic that allows multiple agents to share state and divide work across functional roles.
- Work on context handoff between agents mid-task, so that downstream agents act on current state rather than on decisions that have since been revised.
- Contribute to an early-stage codebase alongside the founding team as the product moves toward pilot deployments.

### Tabner Inc • Recruitment and Client Services Intern

*April 2024 to August 2024*

- Conducted needs discovery sessions with clients and senior consultants, translating stated hiring goals into written job descriptions.
- Sourced and qualified candidates by screening resumes against client requirements, building candidate pipelines for open roles.
- Coordinated interview scheduling end to end, maintaining alignment between clients, candidates and internal recruiters.
- Researched industry hiring trends and presented findings used to refine client recruitment strategy.
- Maintained structured candidate databases, an early form of CRM-style pipeline management.

## TECHNICAL PROJECTS

### F1TENTH Autonomous Racing • Team Developer

*January 2025 | ROS, Python*

Built a 1/10-scale autonomous race car with a multidisciplinary team drawn from software, mechanical and electrical backgrounds.

- Implemented path planning, obstacle avoidance and real-time sensor processing in ROS and Python.
- Tuned PID control and localization algorithms across repeated test-and-failure cycles to improve navigation accuracy on dynamic tracks.
- Integrated LIDAR, IMU and camera hardware with the software stack, translating technical tradeoffs for teammates outside software.

### Route Optimization System

*Python, OR-Tools, Flask, Folium*

Logistics routing tool designed to reduce travel time and cost under real-world operating constraints.

- Implemented and compared Dijkstra's algorithm, A* search and genetic algorithms against constraints including traffic conditions and delivery time windows.
- Built an interactive Flask and Folium interface allowing non-technical stakeholders to test and validate routing solutions without developer involvement.

### 3D Mesh Reconstruction

*Python, PyTorch, PointNet, graph neural networks*

- Contributed to a deep learning pipeline reconstructing 3D meshes from point-cloud input.

### 2D-to-3D Architecture Tool

*Python, machine learning, AutoCAD*

- Helped develop a program converting 2D architectural plans into 3D AutoCAD models, automating a manual step in the design workflow.

### Water Cleaning Bot

*Python, machine learning*

- Designed a low-cost, ML-driven bot for more efficient collection of floating waste.

## LEADERSHIP AND CERTIFICATIONS

### IBM Z Ambassador

*September 2026 to Present*

Selected for IBM's student ambassador program in enterprise computing, a leadership initiative run in partnership with Your Big Year. Completed the IBM Z Xplore Concepts certification as a condition of acceptance.

## TECHNICAL SKILLS

**Programming languages:** Python, Java, SQL

**Frameworks and libraries:** PyTorch, TensorFlow, ROS, Flask, OR-Tools, Folium

**Tools:** Git, AutoCAD, Microsoft Excel

**Domains:** multi-agent systems, machine learning, combinatorial optimization, robotics and control, data analysis, financial analysis

## ADDITIONAL

**Professional competencies:** needs discovery, technical demonstration, presenting technical material to non-technical audiences, cross-disciplinary collaboration.

**Location:** Charlotte, North Carolina.